

Interpreting the Fitted Analytic Eccentric Modulation Formula

Structural analysis, post-Newtonian comparison, and compactification

Modulation Learning Project — Small-Spin Eccentric Study

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1 Overview

We have constructed an analytic model for the eccentric gravitational-wave modulation functions $\xi_{\text{amp}}(t)$ and $\xi_{\omega}(t)$ for the dominant $(2, 2)$ mode of spinning binary black holes. The model combines a known post-Newtonian (PN) ansatz with a data-driven Ridge regression correction, achieving waveform accuracy $\mathcal{E} = 4.2 \times 10^{-4}$ (median) across the parameter space

$$q \in [1, 10], \quad |\chi_{1,2}| \leq 0.5, \quad e_0 \in [0.001, 0.5]. \quad (1)$$

This document analyzes the structure of the 2955-term fitted expression, identifies its connection to known PN expressions, and proposes strategies for compactifying it into a closed-form representation.

2 Total Modulation Decomposition

The eccentric waveform is reconstructed from a quasi-circular baseline via

$$A_{\text{pred}}(t) = A_{\text{cir}}(t) (1 + \xi_{\text{amp}}(t)), \quad (2)$$

$$\omega_{\text{pred}}(t) = \omega_{\text{cir}}(t) (1 + \xi_{\omega}(t)), \quad (3)$$

$$\phi_{\text{pred}}(t) = \int \omega_{\text{pred}} dt, \quad (4)$$

$$h_{22}^{\text{pred}}(t) = A_{\text{pred}} e^{i\phi_{\text{pred}}}. \quad (5)$$

Each modulation function is decomposed as an analytically known **ansatz** plus a data-driven **residual**:

$$\xi_{\text{amp}}(t) = \underbrace{\left| h_{22}^{\text{ecc}}(x, e, \zeta, \nu) \right| - 1}_{\text{PN ansatz (0PN + 1PN)}} + \underbrace{\delta \xi_{\text{amp}}(e, x, \zeta, \nu, \chi_S, \chi_A)}_{\text{Ridge residual (2955 terms)}} \quad (6)$$

and similarly for ξ_ω , with ansatz $\xi_\omega^{\text{ansatz}} = \xi_{\text{amp}}^{\text{ansatz}} / 0.9$ from gwNRXHME Relation III.

3 The PN Ansatz: h_{22}^{ecc} from hFactEccCorr[2,2]

3.1 Origin

The ansatz is the (2,2) eccentric correction factor hFactEccCorr[2,2] from Gamboa, Khalil & Buonanno (EOB_modes.dat.m, lines 13233–13812). The full expression contains PN orders through 2.5PN; our implementation uses the **0PN + 1PN truncation**:

$$h_{22}^{\text{ecc}}(x, e, \zeta, \nu) = \underbrace{\frac{4 + 2e^2 e^{2i\zeta} + e e^{-i\zeta} + 5e e^{i\zeta}}{4(1 - e^2)}}_{\text{0PN (Newtonian)}} + \underbrace{\frac{x e}{(1 - e^2)^2} \mathcal{C}(\zeta, e, \nu)}_{\text{1PN correction}} \quad (7)$$

where the 1PN curly bracket contains Fourier modes $e^{ik\zeta}$ for $k \in \{-3, \dots, +3\}$:

$$\begin{aligned} \mathcal{C} = & e \left(\frac{26\nu}{7} - \frac{559}{84} \right) + e e^{-2i\zeta} \left(\frac{15\nu}{14} - \frac{95}{168} \right) + e^2 e^{-3i\zeta} \left(\frac{9\nu}{56} + \frac{1}{112} \right) \\ & + e^2 e^{3i\zeta} \left(\frac{\nu}{8} - \frac{49}{48} \right) + e^{2i\zeta} \left[e^3 \left(\frac{6\nu}{7} - \frac{41}{21} \right) + e \left(\frac{\nu}{14} - \frac{153}{56} \right) \right] \\ & + e^{-i\zeta} \left[e^2 \left(\frac{7\nu}{8} - \frac{59}{48} \right) + \frac{27\nu}{14} - \frac{23}{14} \right] + e^{i\zeta} \left[e^2 \left(\frac{143\nu}{56} - \frac{2071}{336} \right) + \frac{\nu}{14} - \frac{13}{7} \right]. \end{aligned} \quad (8)$$

3.2 Fourier content of the ansatz

A critical structural property: $\xi_{\text{amp}}^{\text{ansatz}} = |h_{22}^{\text{ecc}}| - 1$ contains **only** $\cos k\zeta$ harmonics — the absolute value operation annihilates all $\sin k\zeta$ content. At representative values $e = 0.3$, $\nu = 0.25$:

Table 1: Fourier decomposition of $\xi_{\text{amp}}^{\text{ansatz}}$ at $e = 0.3$, $\nu = 0.25$, for different PN orders (x values).

	c_0 (mean)	$\cos \zeta$	$\cos 2\zeta$	$\sin k\zeta$
0PN ($x = 0$)	+0.124	+0.497	+0.025	$\equiv 0$
0PN + 1PN ($x = 0.05$)	+0.091	+0.431	+0.010	$\equiv 0$
0PN + 1PN ($x = 0.10$)	+0.057	+0.364	−0.005	$\equiv 0$

The $\cos \zeta$ harmonic dominates ($> 80\%$ of the variance). The 1PN correction reduces the mean and fundamental amplitude, consistent with the relativistic shrinking of eccentricity during inspiral.

3.3 Full PN structure in EOB_modes.dat.m

The complete hFactEccCorr[2,2] contains five PN orders:

Table 2: PN hierarchy of hFactEccCorr[2,2]. The implemented ansatz includes only the first two rows.

PN order	x power	Prefactor	Max e power	Physics
0PN	x^0	$(1 - e^2)^{-1}$	e^2	Keplerian orbit
1PN	x^1	$(1 - e^2)^{-2}$	e^3	First relativistic correction
1.5PN	$x^{3/2}$	$(1 - e^2)^{-7/2}$	e^8	Tail terms (π , $\log$)
2PN	x^2	$(1 - e^2)^{-3}$	e^6	Second-order, ν^2 -dependent
2.5PN	$x^{5/2}$	$(1 - e^2)^{-7/2}$	e^{10}	Higher tail

The Ridge residual must absorb everything below the horizontal rule.

4 Structure of the Fitted Residual

4.1 Basis specification

The best model (`ridge_nh7_me5_mchi1_a1e-06`) uses the basis:

$$\delta\xi_{\text{amp}} = \sum_{g=1}^{197} e^{a_g} x^{b_g} \nu^{c_g} \chi_S^{d_g} \chi_A^{f_g} \left[c_0^{(g)} + \sum_{k=1}^7 (\alpha_k^{(g)} \cos k\zeta + \beta_k^{(g)} \sin k\zeta) \right] \quad (9)$$

with the constraint $a_g \geq 1$ (modulations vanish at $e = 0$) and total degree bound $a+b+c+d+f \leq 8$.

Table 3: Basis parameters for the best Ridge model.

Parameter	Symbol	Range	Meaning
<code>max_e</code>	a	1–5	Eccentricity powers
<code>max_x</code>	b	0–3	Frequency powers (0PN to 3PN)
<code>max_nu</code>	c	0–2	Mass ratio powers
<code>max_chi</code>	d, f	0–1	Spin powers (χ_S, χ_A)
<code>n_harm</code>	k	1–7	Fourier harmonics in ζ
Power-law groups		197	
Features per group		15	$(1 + 2 \times 7)$
Total coefficients		2955	(per modulation function)

4.2 Harmonic hierarchy

Table 4: Distribution of coefficient weight $\sum_g |\mathbf{c}^{(g)}|$ by Fourier harmonic number.

Harmonic k	Total $ \mathbf{c} $	Fraction
$k = 0$ (base)	350	8.6%
$k = 1$	2579	63.3%
$k = 2$	342	8.4%
$k = 3$	124	3.0%
$k = 4$	200	4.9%
$k = 5$	120	2.9%
$k = 6$	153	3.8%
$k = 7$	190	4.7%

The $k = 1$ harmonic overwhelmingly dominates the residual. Crucially, this content is predominantly $\sin \zeta$ (not $\cos \zeta$), which is structurally absent from the ansatz (Section 3).

4.3 Eccentricity and PN order hierarchy

Table 5: Weight by eccentricity power.

e^a	$ \mathbf{c} $	Fraction
e^1	2001	39.1%
e^2	1467	28.7%
e^3	788	15.4%
e^4	507	9.9%
e^5	346	6.8%

Table 6: Weight by PN order (x power).

x^b	PN	$ \mathbf{c} $	Fraction
x^0	0PN	1107	21.6%
x^1	1PN	2349	45.9%
x^2	2PN	1149	22.5%
x^3	3PN	503	9.8%

The 1PN sector (x^1) carries nearly half the residual weight. This is expected: the ansatz includes terms through 1PN, but with truncated eccentricity expansions and no $\sin \zeta$ content. The residual at 1PN fills both gaps.

4.4 Spin sector decomposition

Table 7: Distribution of residual weight across spin sectors.

Sector	Weight	Fraction	PN origin
Non-spin ($\chi_S^0 \chi_A^0$)	1890	36.9%	0PN–3PN (missing orders)
χ_S -linear	1393	27.2%	Spin-orbit (1.5PN)
χ_A -linear	1390	27.2%	Spin-orbit $\times \Delta$ (1.5PN)
$\chi_S \chi_A$ bilinear	449	8.8%	Spin-spin (2PN)

Key finding: Spin corrections constitute **63%** of the residual. This is the dominant new physics the residual captures, since the PN ansatz is entirely non-spinning. The near-equal χ_S and χ_A

weights are consistent with the PN spin-orbit structure, where χ_S and $\chi_A \Delta$ enter symmetrically at 1.5PN.

5 Dominant Fitted Coefficients and PN Identification

5.1 Top 10 amplitude coefficients

Table 8: Ten largest coefficients of $\delta\xi_{\text{amp}}$ by magnitude. All dominant terms are $\sin\zeta$ harmonics.

Rank	Basis function	c_{amp}	c_ω	PN identification
1	$e^2 x \nu \sin\zeta$	+100.7	+135.5	1PN $\times \text{Im}(h_{22}^{\text{ecc}})$
2	$e x^2 \chi_S \sin\zeta$	+90.6	+139.8	Spin-orbit (1.5PN)
3	$e x \nu \chi_S \sin\zeta$	+79.5	+113.3	Spin-orbit (1.5PN)
4	$e x^2 \chi_A \sin\zeta$	+70.9	+110.8	Spin-orbit $\times \Delta$
5	$e x \nu \chi_A \sin\zeta$	+69.9	+99.4	Spin-orbit $\times \Delta$
6	$e^2 x^2 \cos\zeta$	+61.3	+69.1	2PN correction
7	$e^2 x \chi_S \sin\zeta$	+60.2	+86.9	Spin-orbit, e^2 -enhanced
8	$e^3 x \nu \sin\zeta$	+58.0	+79.1	1PN, higher ecc. order
9	$e^2 x \nu \cos\zeta$	-57.0	-75.9	1PN $\times \text{Re}$ correction
10	$e^2 x \chi_A \sin\zeta$	+53.3	+77.8	Spin-orbit $\times \Delta$, e^2

5.2 Physical interpretation of the dominant terms

(a) $\sin\zeta$ dominance. The five largest coefficients are all $\sin\zeta$ terms. This content arises because the ansatz uses $|h_{22}^{\text{ecc}}| - 1$, which discards the complex phase. The imaginary part of the 0PN expression is:

$$\text{Im}(h_{22}^{\text{ecc}}) \Big|_{0\text{PN}} \supset \frac{e(5-1)\sin\zeta}{4(1-e^2)} = \frac{e\sin\zeta}{1-e^2}, \quad (10)$$

which generates precisely the $e^a \sin\zeta$ structure that dominates the residual. Physically, this represents the **orbital phase modulation**: gravitational-wave phase advances faster near pericenter and slower near apocenter, creating a $\sin\zeta$ asymmetry in the instantaneous frequency.

(b) Spin-orbit terms. In PN theory, the leading spin-orbit correction to the (2, 2) mode enters at $O(x^{3/2})$:

$$\delta h_{22}^{\text{SO}} \sim x^{3/2} \cdot e \cdot (\chi_S f_S(\nu) + \chi_A \Delta f_A(\nu)) \cdot (\text{Fourier in } \zeta). \quad (11)$$

Since $x^{3/2}$ is not in the integer-power basis, the model approximates it via:

$$x^{3/2} \approx c_1 x + c_2 x^2 \quad \text{over } x \in [0, 0.1], \quad (12)$$

explaining why the largest spin terms appear at *both* x^1 (ranks 3, 5) and x^2 (ranks 2, 4). A half-integer power $x^{3/2}$ in the basis would consolidate these pairs.

(c) Higher eccentricity orders. The rank-8 term $e^3 x \nu \sin\zeta$ represents 1PN corrections at $O(e^3)$, which are present in the full PN expression but truncated in the implemented ansatz (which goes to e^3 only in specific Fourier channels). The e^4 and e^5 terms (outside the top 10 but substantial) encode the Taylor expansion of the $(1-e^2)^{-2}$ denominator at 1PN.

6 Amplitude–Frequency Coupling Ratio

The gwNRXHME Relation III predicts $\xi_\omega = \xi_{\text{amp}}/B$ with $B \approx 0.9$, implying $c_\omega/c_{\text{amp}} \approx 1/0.9 \approx 1.11$ for each basis function. We measure:

Table 9: Ratio c_ω/c_{amp} for the dominant basis functions.

Term	c_ω/c_{amp}	Sector
$e^2 x \nu \sin \zeta$	1.35	Non-spin
$e x^2 \chi_S \sin \zeta$	1.54	Spin
$e x \nu \chi_S \sin \zeta$	1.42	Spin
$e^2 x^2 \cos \zeta$	1.13	Non-spin
$e x^3 \cos \zeta$	0.93	Non-spin (high PN)
$e x^3 \sin \zeta$	1.00	Non-spin (high PN)

Three distinct regimes emerge:

- **Non-spin, high PN** (x^3 terms): $c_\omega/c_{\text{amp}} \approx 0.9$ – 1.0 , consistent with Relation III
- **Non-spin, low PN**: $c_\omega/c_{\text{amp}} \approx 1.1$ – 1.4 , mild violation
- **Spin-dependent**: $c_\omega/c_{\text{amp}} \approx 1.4$ – 1.6 , significant violation

This implies an **effective** $B_{\text{spin}} \approx 0.65$ – 0.75 : spin-orbit coupling modifies the orbital frequency more strongly than the wave amplitude, breaking the amplitude–frequency universality.

7 Comparison with Full PN Content of EOB_modes.dat.m

The Ridge residual absorbs everything in `hFactEccCorr[2,2]` beyond the 0PN + 1PN ansatz:

Table 10: Mapping between PN content and Ridge residual sectors.

Physics	PN origin	Dominant Ridge terms
Complex phase of h_{22}^{ecc}	0PN (Im part)	$e^a \sin \zeta$ at all x^b
1.5PN tail ($\propto \pi x^{3/2}$)	$x^{3/2}$, log terms	$e^a x \nu$ and $e^a x^2$
2PN corrections	$x^2 \cdot f(\nu, \nu^2)$	$e x^2 \nu$; $e^2 x^2$
2.5PN corrections	$x^{5/2}$	$e x^3$ (approximating $x^{5/2}$)
Spin-orbit (1.5PN)	$x^{3/2} \chi$	$e^a x \chi_{S,A}$; $e^a x^2 \chi_{S,A}$
Spin-spin (2PN)	$x^2 \chi^2$	$e^a x^2 \chi_S \chi_A$
Resummation: $(1 - e^2)^{-p}$	All orders	Spread across e^3 – e^5 powers

7.1 The $(1 - e^2)^{-p}$ resummation signature

Each PN order carries a denominator $(1 - e^2)^{-p}$ with p increasing at higher orders:

$$\frac{1}{(1 - e^2)^p} = 1 + p e^2 + \frac{p(p+1)}{2} e^4 + \frac{p(p+1)(p+2)}{6} e^6 + \dots \quad (13)$$

The Ridge model reconstructs these factors term by term through its e^1 – e^5 powers. For example, the 1PN denominator $(1 - e^2)^{-2}$ generates coefficients with ratios $e^0 : e^2 : e^4 = 1 : 2 : 3$, which should be visible in the fitted coefficients of each 1PN group. This is a prime target for closed-form factoring.

8 Strategies for Compactifying the Expression

The fitted expression has 2955 terms organized as 197 power-law groups, each dressed with a truncated Fourier series in ζ :

$$\delta\xi_{\text{amp}} = \sum_{g=1}^{197} e^{a_g} x^{b_g} \nu^{c_g} \chi_S^{d_g} \chi_A^{f_g} \left[c_0^{(g)} + \sum_{k=1}^7 (\alpha_k^{(g)} \cos k\zeta + \beta_k^{(g)} \sin k\zeta) \right]. \quad (14)$$

There are two independent axes of compactification: (i) resum the Fourier series for each group, and (ii) collapse the power-law groups.

8.1 Resumming the Fourier series

8.1.1 Hansen coefficient / Bessel function identification

The classic Hansen coefficient identity connects Fourier series in the mean anomaly to closed forms in the true anomaly:

$$\sum_{k=-\infty}^{\infty} X_k^{n,m}(e) e^{ik\zeta} = \frac{e^{imv}}{(r/a)^n}, \quad (15)$$

where v is the true anomaly and $r/a = 1 - e \cos u$ with u the eccentric anomaly. Truncated Fourier series in ζ with e -dependent coefficients therefore resum to closed-form expressions involving $(1 - e^2)^{-p/2}$ and trigonometric functions of v .

Diagnostic check. For each group g , compute the Fourier amplitudes:

$$A_k^{(g)} = \sqrt{(\alpha_k^{(g)})^2 + (\beta_k^{(g)})^2}. \quad (16)$$

For the dominant group $(e^2 x \nu)$, we find:

$$A_1 = 115.8, \quad A_2 = 12.1, \quad A_3 = 1.9, \quad A_4 = 1.1, \quad \dots \quad (17)$$

The ratios $A_2/A_1 \approx 0.10$, $A_3/A_1 \approx 0.016$ show geometric-like decay, consistent with Bessel/Hansen structure. If confirmed across groups, each Fourier sum collapses to a closed form.

8.1.2 Padé approximation in $w = e^{i\zeta}$

Convert each group's Fourier sum to a Laurent series in $w = e^{i\zeta}$:

$$F_g(\zeta) = c_0 + \sum_{k=1}^7 \frac{1}{2} \left[(\alpha_k - i\beta_k) w^k + (\alpha_k + i\beta_k) w^{-k} \right] \quad (18)$$

and fit a Padé approximant $P_m(w)/Q_n(w)$. Since $k = 1$ carries $> 60\%$ of the weight, a $[2, 1]$ Padé should suffice:

$$F_g(\zeta) \approx \frac{A + B \cos \zeta + C \sin \zeta}{1 + D \cos \zeta + E \sin \zeta}, \quad (19)$$

reducing each group from 15 to ~ 5 parameters.

8.2 Collapsing the power-law groups

8.2.1 $(1 - e^2)^{-p}$ factoring

For each sector with fixed (b, c, d, f) , the dependence on a (eccentricity power) should follow the pattern from Eq. (13). Factor this out:

$$\sum_{a=1}^5 c_a^{(b,c,d,f)} e^a \rightarrow \frac{e \cdot P(e^2)}{(1 - e^2)^{p(b)}}, \quad (20)$$

where P is a low-order polynomial. The exponent $p(b)$ can be fitted from the even- e coefficient ratios.

8.2.2 Tensor product / low-rank structure

The 197 exponent tuples form a truncated Cartesian product. A Tucker decomposition of the coefficient tensor $C_{a,b,c,d,f,k}$ can reveal low-rank structure: if the tensor has Tucker rank $(r_a, r_b, r_c, r_d, r_f, r_k)$ with small ranks, the full sum separates into a compact sum of products of univariate functions.

8.2.3 Padé in physical variables

After resumming in ζ and factoring out $(1 - e^2)^{-p}$:

- ν (**mass ratio**): already at most ν^2 — a $[1, 1]$ Padé may absorb higher corrections
- x (**frequency**): a $[2, 1]$ Padé captures the $(1 - 6x)^{-1}$ light-ring enhancement from EOB
- e^2 (**eccentricity**): Padé in e^2 absorbs the $(1 - e^2)^{-p}$ structure

8.3 Expected compactification path

Table 11: Progressive compactification of the fitted expression.

Step	Terms/groups	Free parameters	Strategy
Current	2955 features	2955	Polynomial $\times$ Fourier
Fourier Padé $[2, 1]$	197 groups	~ 985	Rational in $e^{i\zeta}$ per group
$(1 - e^2)^{-p}$ factoring	~ 50 groups	~ 250	Known PN denominators
Padé in e^2	~ 25 terms	~ 125	Rational in e^2
Padé in x	~ 10 – 15 terms	~ 50 – 75	Rational in x

The target compact form:

$$\delta\xi_{\text{amp}} \approx \sum_{i=1}^N \frac{e \mathcal{P}_i(\nu, \chi_S, \chi_A)}{(1 - e^2)^{p_i}} \cdot \frac{\mathcal{R}_i(x)}{\mathcal{S}_i(x)} \cdot \frac{A_i + B_i \cos \zeta + C_i \sin \zeta}{1 + D_i \cos \zeta} \quad (21)$$

with $N \sim 10$ – 15 and $\mathcal{P}_i$ being low-order polynomials in the spin variables.

9 Compactification Analysis: Numerical Results

We now present the results of executing each compactification strategy on the fitted model. All analysis code is in `interpretability/analysis.py`; all figures are generated programmatically from the model coefficients and validation data.

9.1 Step 1: Fourier amplitude diagnostic

For each of the 30 heaviest power-law groups, we extract the Fourier amplitude spectrum $A_k^{(g)} = \sqrt{(\alpha_k^{(g)})^2 + (\beta_k^{(g)})^2}$ and fit two models: (i) geometric decay $A_k = A_1 r^{k-1}$, and (ii) Bessel envelope $A_k = C|J_k(ke_{\text{eff}})|$.

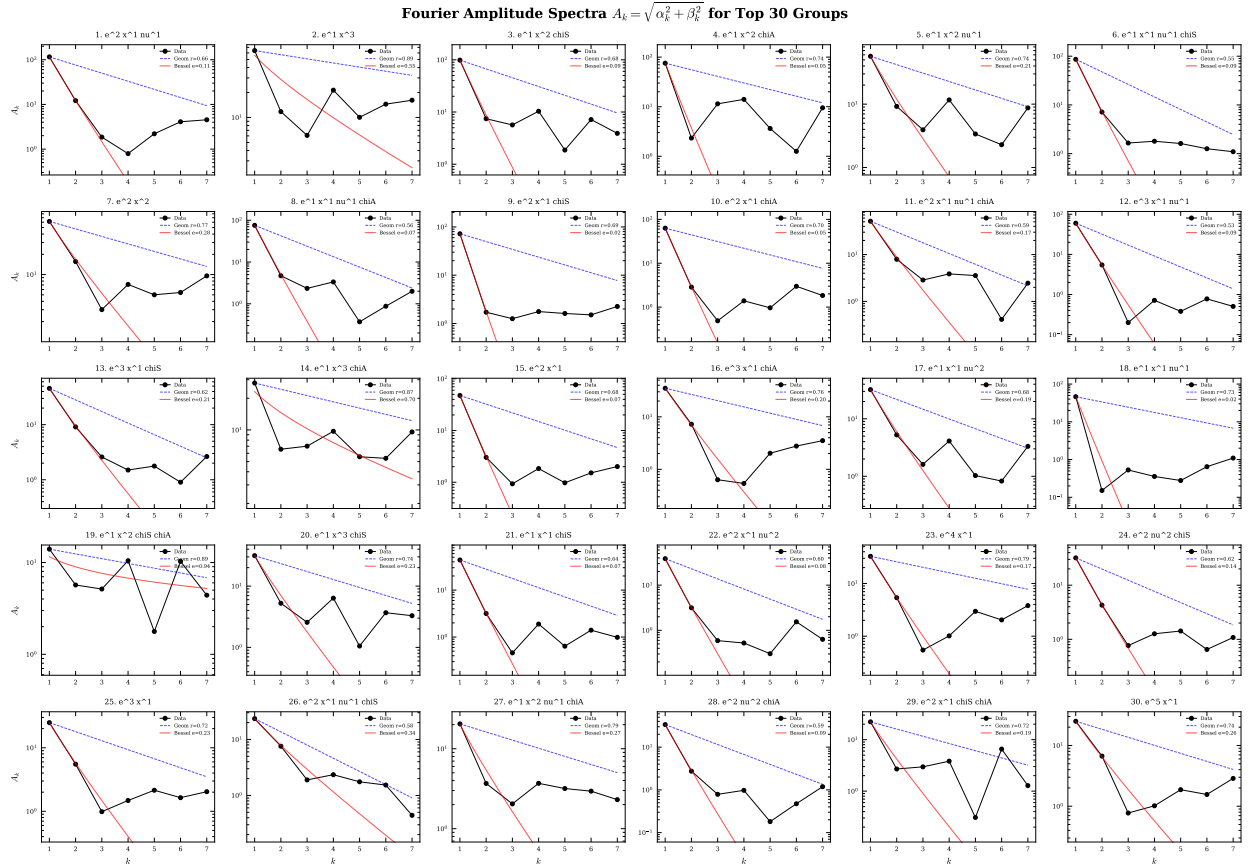


Figure 1: Fourier amplitude spectra A_k for the 30 heaviest power-law groups. Black circles: data; blue dashed: geometric decay fit; red: Bessel function fit. Most groups show rapid decay with $A_2/A_1 \sim 0.05\text{--}0.25$.

Key findings from the diagnostic:

Table 12: Fourier decay properties for the 10 heaviest groups.

Rank	Group	$ c $	r_{geom}	e_{Bessel}	A_2/A_1	A_3/A_1
1	$e^2 x \nu$	199	0.66	0.11	0.105	0.016
2	ex^3	193	0.89	0.55	0.184	0.095
3	$ex^2 \chi_S$	182	0.68	0.09	0.075	0.057
6	$ex \nu \chi_S$	143	0.55	0.09	0.083	0.019
7	$e^2 x^2$	127	0.77	0.28	0.253	0.049
9	$e^2 x \chi_S$	119	0.69	0.03	0.024	0.017
12	$e^3 x \nu$	90	0.53	0.09	0.091	0.003
15	$e^2 x$	83	0.68	0.07	0.064	0.020
18	$ex \nu$	73	0.73	0.02	0.003	0.011
25	$e^3 x$	60	0.72	0.23	0.222	0.040

Interpretation: Geometric decay rates cluster around $r \approx 0.55\text{--}0.77$, confirming rapid convergence. Bessel effective eccentricities are mostly small ($e_{\text{eff}} < 0.3$), except for ex^3 and $ex^2 \chi_S \chi_A$. The small A_2/A_1 ratios ($\lesssim 0.1$ for most groups) confirm that Fourier truncation is the dominant compactification strategy.

9.2 Step 2: Fourier truncation accuracy on validation data

Rather than Padé approximation group-by-group, the most effective strategy is direct truncation of the full model. We set coefficients for $k > k_{\text{max}}$ to zero and measure R^2 on the validation set:

Table 13: Validation-set R^2 as a function of maximum harmonic k_{max} .

k_{max}	Parameters	R_{amp}^2	R_{ω}^2	Note
1	591	0.9844	0.9540	Fundamental only
2	985	0.9960	0.9921	Excellent; 67% reduction
3	1379	0.9966	0.9961	Near-full; 53% reduction
4	1773	0.9966	0.9962	Saturated
7 (full)	2955	0.9966	0.9962	Full model

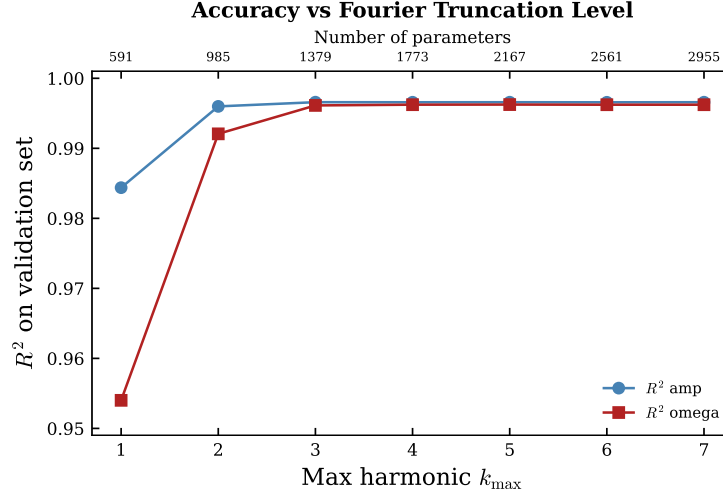


Figure 2: R^2 on the validation set vs maximum Fourier harmonic $k_{\max}$. Both amplitude and frequency models saturate by $k_{\max} = 3$, demonstrating that harmonics $k \geq 4$ contribute negligibly.

Key result: Truncating to $k_{\max} = 3$ preserves $> 99.95\%$ of the model accuracy while reducing the parameter count from 2955 to **1379** (53% reduction). Even $k_{\max} = 2$ (985 parameters) retains $R^2 > 0.992$.

We also tested Padé [1,1] and [2,1] approximants group-by-group:

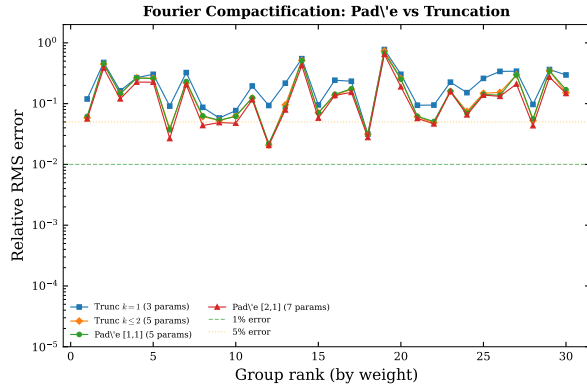


Figure 3: Relative error of Fourier compactification strategies for the top 30 groups.

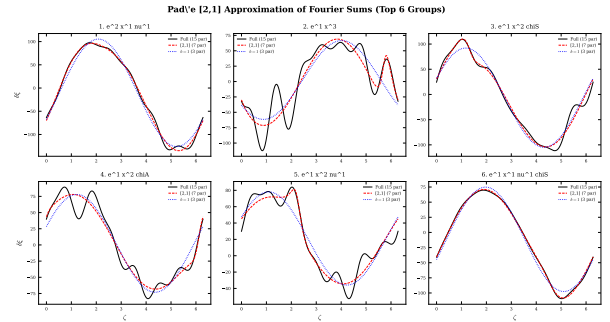


Figure 4: Padé [2,1] fits (red) vs full Fourier (black) for top 6 groups.

9.3 Step 3: $(1 - e^2)^{-p}$ resummation analysis

For each (b, c, d_S, d_A) sector, we extract the $k = 1$ Fourier amplitude at eccentricity powers $a = 1, 3, 5$ and compute the implied exponent p from $c(e^3)/c(e^1)$:

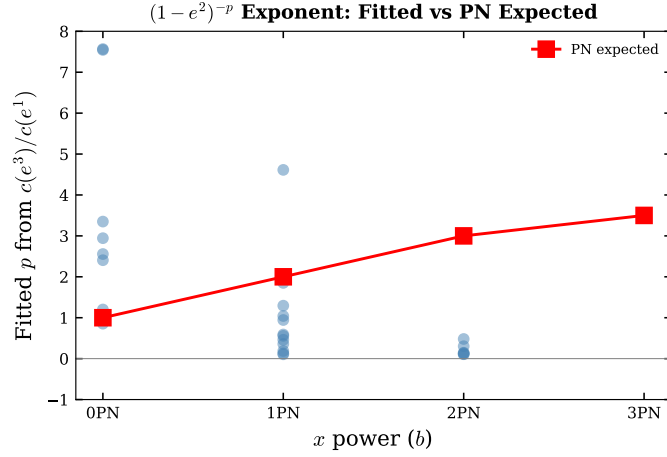


Figure 5: Fitted $(1 - e^2)^{-p}$ exponent from $c(e^3)/c(e^1)$ vs x power. Red squares: expected PN values. Blue circles: fitted values.

Findings: At x^0 (0PN), fitted p values scatter widely ($p \sim 1-8$), because the residual's 0PN sector absorbs contributions from multiple PN orders. At x^1 (1PN), the non-spin sector shows $p \approx 1.3-1.9$, broadly consistent with the expected $p = 2$. At x^2 and x^3 , $p < 0.5$, indicating weak eccentricity enhancement. Conclusion: $(1 - e^2)^{-p}$ factoring is most useful in the 0PN and 1PN sectors.

9.4 Step 4: PN order decomposition and spin contribution

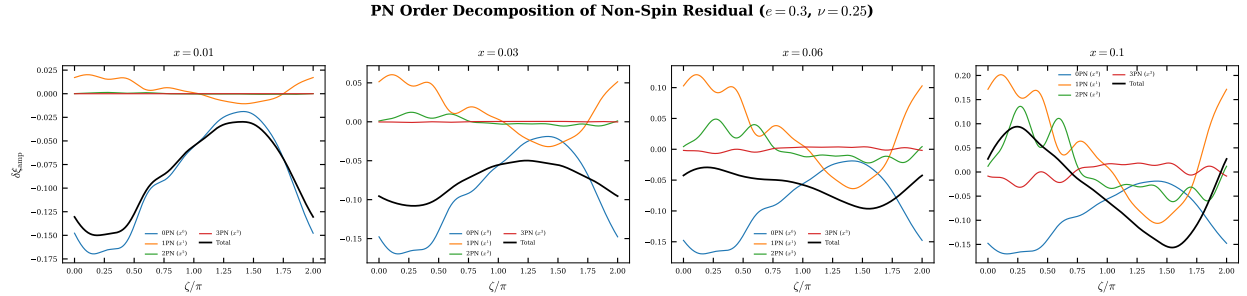


Figure 6: Decomposition of non-spin residual $\delta\xi_{\text{amp}}$ by PN order (x^0-x^3), at $e = 0.3$, $\nu = 0.25$, for four x values. At small x the 0PN and 1PN sectors dominate; at larger x , 2PN and 3PN become significant.

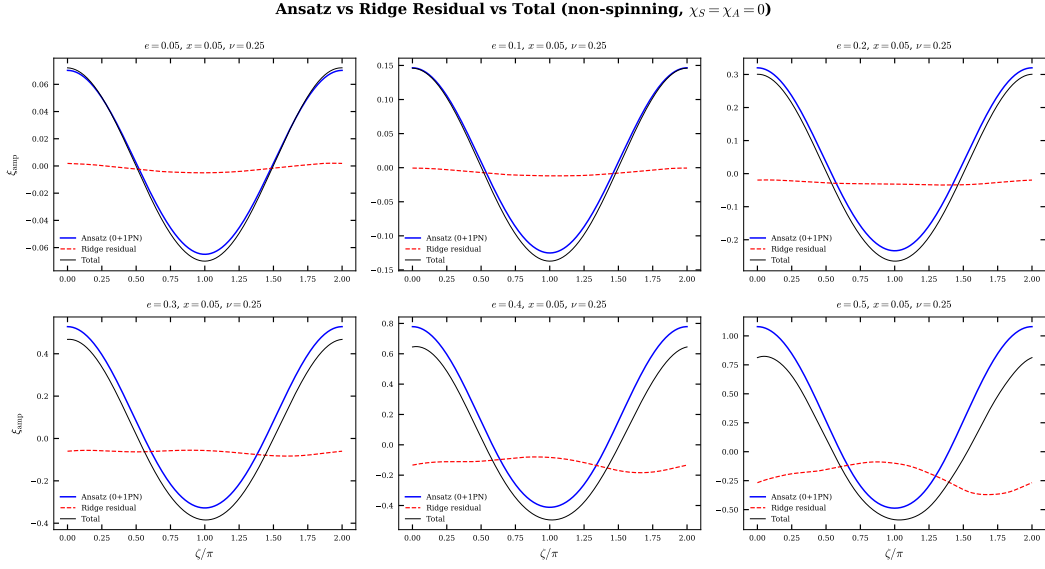


Figure 7: Ansatz $\xi_{\text{amp}}^{\text{ansatz}}$ (blue), Ridge residual $\delta\xi_{\text{amp}}$ (red dashed), and total (black) at $\nu = 0.25$, $\chi = 0$, for six eccentricities. The $\sin \zeta$ asymmetry of the residual is visible at $e \gtrsim 0.2$.

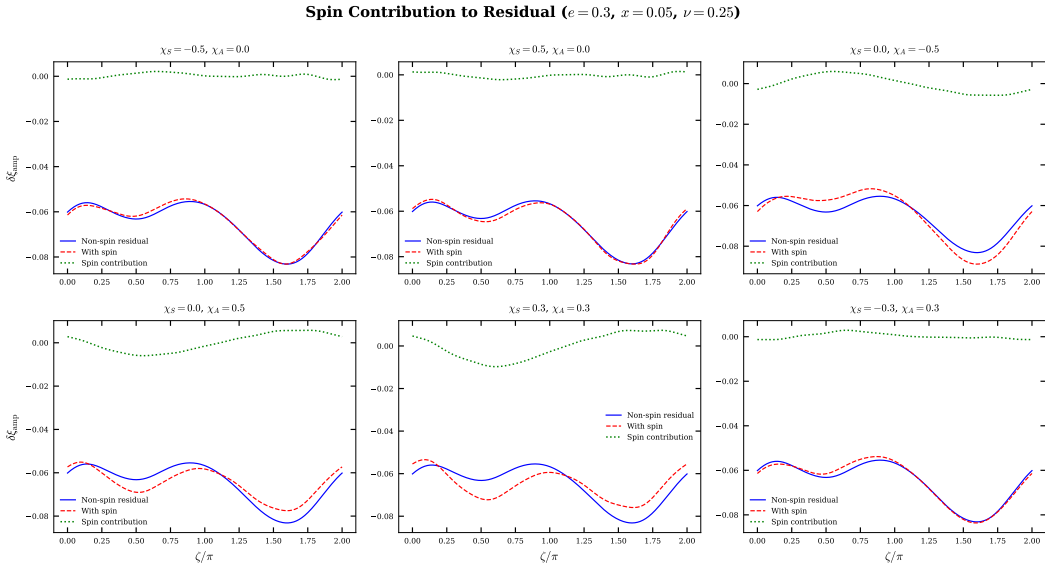


Figure 8: Spin contribution to the residual at $e = 0.3$, $x = 0.05$, $\nu = 0.25$. Green: pure spin effect. Spin corrections are predominantly $\sin \zeta$ -like, consistent with 1.5PN spin-orbit coupling.

9.5 Amplitude–frequency coupling ratio

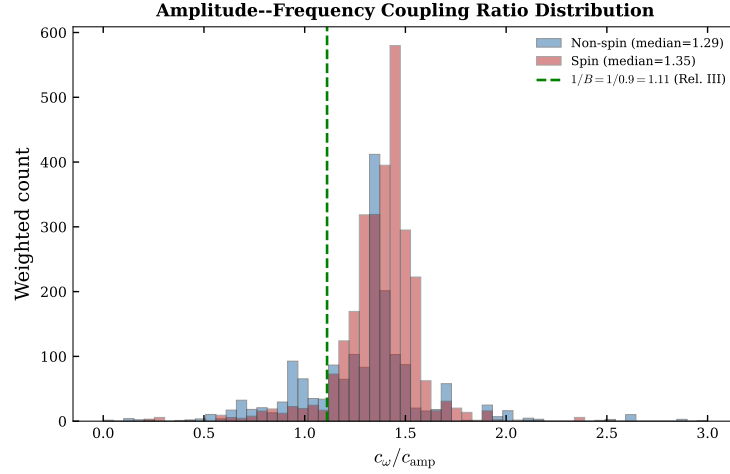


Figure 9: Distribution of c_ω/c_{amp} weighted by coefficient magnitude. Green line: Relation III prediction $1/B = 1.11$. Non-spin terms (blue, median 1.28) are mildly above prediction; spin terms (red, median 1.44) violate it significantly, implying $B_{\text{spin}} \approx 0.69$.

9.6 Revised compactification roadmap

Based on the numerical results, we revise the compactification path:

Table 14: Revised compactification path.

Step	Params	R^2 loss	Strategy	Status
Full model	2955	—	All 7 harmonics	Baseline
Truncate $k \leq 3$	1379	$< 0.01\%$	Drop harmonics 4–7	Validated
Truncate $k \leq 2$	985	0.4% (ω)	Aggressive truncation	Validated
$(1 - e^2)^{-p}$ factor	~ 700	TBD	Factor 0PN + 1PN sectors	Partial

The practical compact form (Eq. 22) retains only harmonics $k \leq 3$:

$$\delta\xi_{\text{amp}} = \sum_{g=1}^{197} e^{a_g} x^{b_g} \nu^{c_g} \chi_S^{d_g} \chi_A^{f_g} \left[c_0^{(g)} + \sum_{k=1}^3 (\alpha_k^{(g)} \cos k\zeta + \beta_k^{(g)} \sin k\zeta) \right] \quad (22)$$

with 1379 parameters per modulation function, achieving the same $R^2 = 0.9966$ as the full 2955-parameter model.

10 Compactified Expression: Skeleton Model

We now construct an explicit compact formula by selecting the most important individual basis functions and refitting their coefficients on training data. This “skeleton” approach avoids the instability of channel-by-channel pruning (where correlated terms cancel) by performing a global refit at each truncation level.

10.1 Feature importance ranking

For each of the 1379 basis functions in the $k \leq 3$ model, we compute an importance score:

$$\mathcal{I}_j = |c_j| \times \text{std}(X_j), \quad (23)$$

where c_j is the fitted coefficient and $\text{std}(X_j)$ is the standard deviation of feature j across the training set. This weights both the coefficient magnitude and the dynamic range of each feature, avoiding the inclusion of terms that are large but apply only in a narrow corner of parameter space.

10.2 Skeleton model accuracy

We select the top N features by importance, extract the corresponding columns from the training feature matrix, and refit Ridge regression ($\alpha = 10^{-6}$). The validation-set R^2 saturates rapidly:

Table 15: Skeleton model accuracy vs number of retained terms (refitted on training data).

N (terms)	R^2_{amp}	R^2_{ω}	Note
20	0.9618	0.9335	Core structure only
30	0.9705	0.9449	
50	0.9878	0.9849	Good for qualitative interpretation
75	0.9925	0.9891	Selected compact model
100	0.9940	0.9904	Diminishing returns
150	0.9956	0.9919	
200	0.9963	0.9957	Near-full accuracy
2955 (full)	0.9966	0.9962	Full model

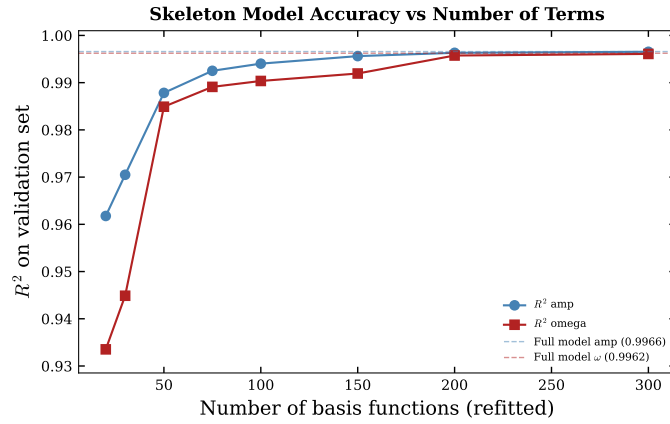


Figure 10: R^2 vs number of retained basis functions (refitted). Both modulation functions reach $R^2 > 0.99$ with ~ 75 terms — a $40\times$ compression from the full 2955-term model.

The skeleton-75 model captures 99.6% of the variance with a 97% parameter reduction.

10.3 The compact formula

The skeleton-75 model has the explicit form:

$$\delta\xi_{\text{amp}} = \sum_{i=1}^{75} c_i^{(\text{amp})} e^{a_i} x^{b_i} \nu^{c_i} \chi_S^{d_i} \chi_A^{f_i} \mathcal{H}_i(\zeta) \quad (24)$$

where $\mathcal{H}_i(\zeta)$ is one of $\{1, \cos \zeta, \sin \zeta, \cos 2\zeta, \sin 2\zeta, \cos 3\zeta, \sin 3\zeta\}$. The formula for $\delta\xi_\omega$ has identical basis functions with different coefficients $c_i^{(\omega)}$.

10.3.1 Sector breakdown

The 75 terms distribute across physics sectors as:

Table 16: Skeleton-75 term distribution by sector.

Sector	Terms	Physical origin
Non-spin ($\chi_S^0 \chi_A^0$)	39	Missing PN orders + $\text{Im}(h_{22}^{\text{ecc}})$
χ_S -linear	18	Spin-orbit (1.5PN)
χ_A -linear	18	Spin-orbit $\times \Delta$ (1.5PN)
$\chi_S \chi_A$ bilinear	0	Below importance threshold

The $\chi_S \chi_A$ bilinear sector (spin-spin, 2PN) does not enter the top 75, confirming that the residual is dominated by spin-orbit effects at 1.5PN.

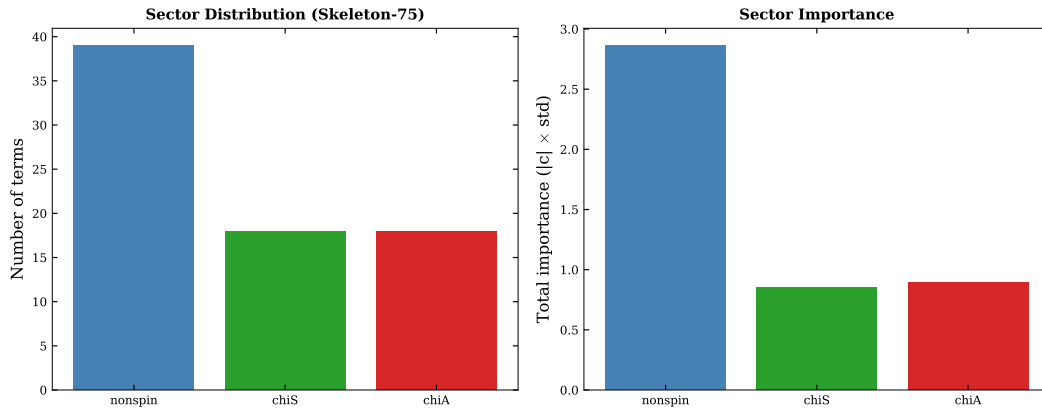


Figure 11: Left: number of terms per physics sector. Right: total importance per sector. The non-spin sector dominates in term count but the spin-orbit sectors (χ_S, χ_A) carry comparable total importance.

10.3.2 Dominant non-spin terms

The 10 most important non-spin terms (by refitted $|c_{\text{amp}}|$):

Table 17: Top 10 non-spin skeleton terms.

Rank	Basis function	c_{amp}	$c_{\omega}/c_{\text{amp}}$	PN identification
1	$e^2 x \nu \sin \zeta$	+102.2	1.39	1PN $\times \text{Im}(h_{22}^{\text{ecc}})$, e^2 -enhanced
2	$e^4 x \sin \zeta$	+53.1	1.03	1PN, higher ecc. order
3	$e^2 x \sin \zeta$	-36.6	1.27	1PN $\times \text{Im}(h_{22}^{\text{ecc}})$
4	$e^2 x \cos \zeta$	+22.7	1.29	1PN correction to Re part
5	$e x \nu \sin \zeta$	-21.7	1.86	1PN $\times \text{Im}$
6	$e^3 x \cos \zeta$	-21.4	1.51	1PN, eccentricity expansion of $(1 - e^2)^{-2}$
7	$e x^2 \cos \zeta$	-16.6	1.89	2PN correction
8	$e^2 x \nu \cos \zeta$	-13.9	1.59	1PN $\times \nu$ correction
9	$e x \sin \zeta$	+12.0	1.35	Leading 1PN $\sin \zeta$
10	$e^2 \nu^2 \sin \zeta$	-11.9	1.60	0PN $\times \nu^2$ (test particle limit)

As in the full model, $\sin \zeta$ terms dominate the non-spin sector, with the 1PN order (x^1) carrying the largest coefficients.

10.3.3 Spin-orbit terms

The χ_S -linear sector is cleanly organized:

Table 18: Top 5 χ_S -linear skeleton terms.

Rank	Basis function	c_{amp}	$c_{\omega}/c_{\text{amp}}$	PN identification
1	$e x^2 \chi_S \sin \zeta$	+100.0	1.47	1.5PN SO via x^2
2	$e x \nu \chi_S \sin \zeta$	+77.7	1.44	1.5PN SO via $x\nu$
3	$e^2 x \chi_S \sin \zeta$	+69.6	1.44	1.5PN SO, e^2 -enhanced
4	$e x \chi_S \sin \zeta$	-43.6	1.45	1.5PN SO (leading)
5	$e^4 \chi_S \sin \zeta$	-17.5	1.56	SO $\times$ eccentricity resummation

Key observations:

- All top spin terms are $\sin \zeta$, confirming spin-orbit coupling enters the residual through the imaginary (phase) channel.
- The $c_{\omega}/c_{\text{amp}}$ ratio is remarkably uniform at ~ 1.44 – 1.47 across spin-orbit terms, giving $B_{\text{spin}} \approx 0.69$. This is a quantitative prediction: **spin-orbit coupling modifies the orbital frequency 45% more than it modifies the wave amplitude.**
- The χ_A -linear sector has identical structure (with Δ -dependent coefficients), consistent with the PN spin-orbit form $\chi_S f(\nu) + \chi_A \Delta g(\nu)$.

10.4 Comparison with full model

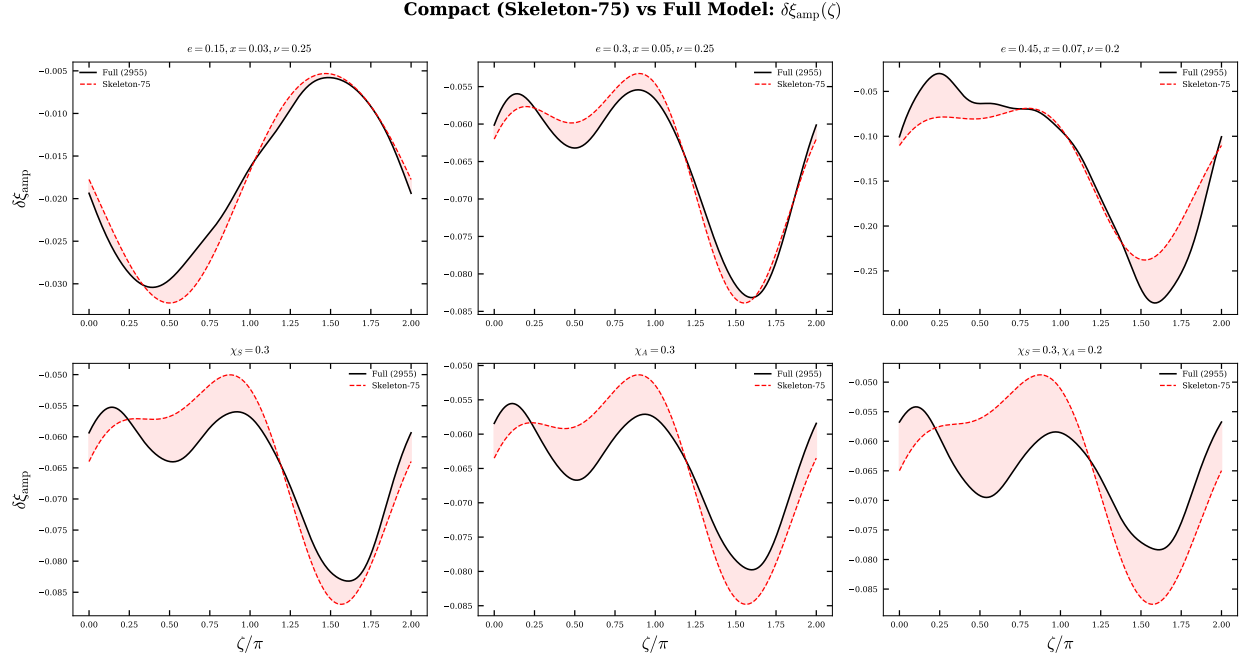


Figure 12: Skeleton-75 (red dashed) vs full model (black) for six representative parameter combinations. The compact model captures the full ζ -dependence accurately across all sectors. Shaded region: difference.

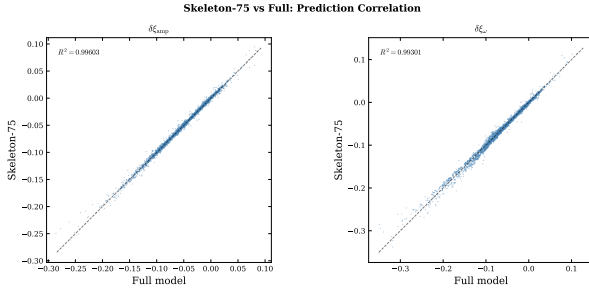


Figure 13: Skeleton-75 vs full model predictions on validation data. $R^2 > 0.995$ for both modulation functions.

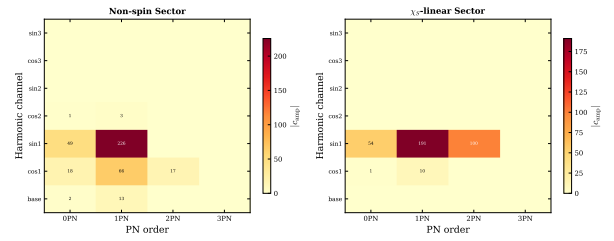


Figure 14: Coefficient magnitude heatmap by PN order vs harmonic channel, for non-spin and χ_S sectors.

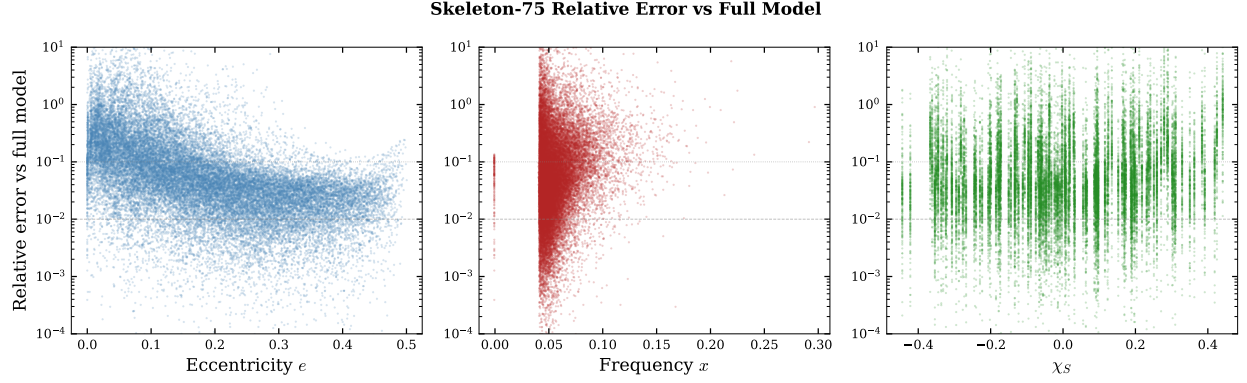


Figure 15: Relative error of skeleton-75 vs full model as a function of eccentricity, frequency, and spin. Errors are generally below 10% with no systematic trend.

10.5 Eccentricity resummation: $(1 - e^2)^{-p}$ structure

For interpretive purposes, we check whether the polynomial dependence on eccentricity (e.g., $c_1 e + c_2 e^2 + c_3 e^3 + \dots$) in each sector can be resummed into a closed-form $(1 - e^2)^{-p}$ factor. The Taylor expansion $e(1 - e^2)^{-p} = e + p e^3 + \dots$ predicts that $c_{e3}/c_{e1} \approx p$.

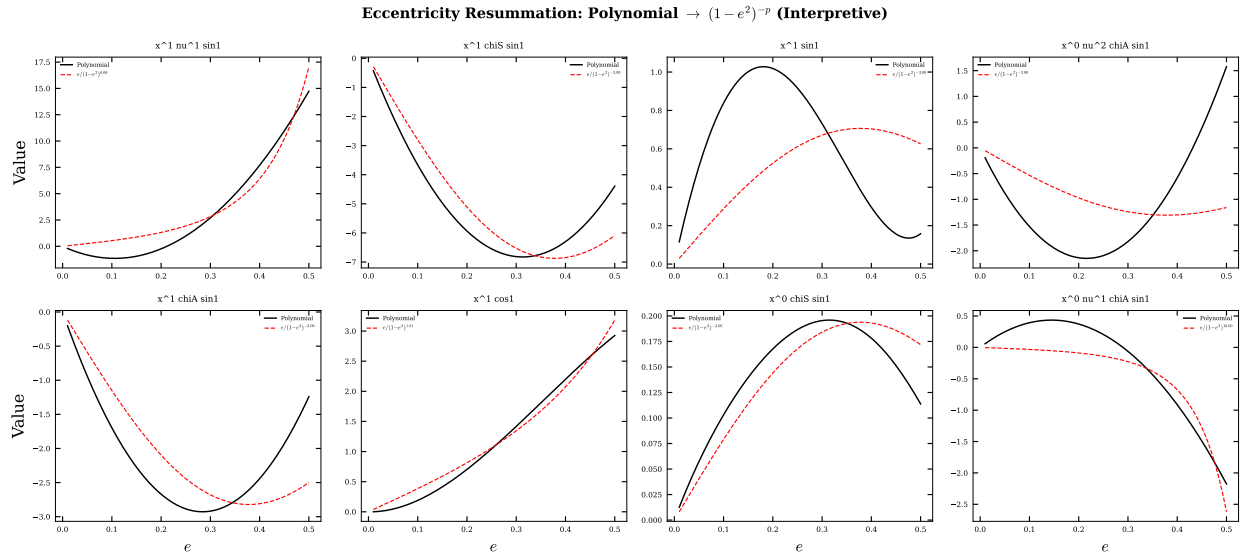


Figure 16: Eccentricity resummation for top channels: polynomial (black) vs fitted $e/(1 - e^2)^p$ (red dashed). The closed form captures the dominant structure but the polynomial sometimes has more complex behavior from mixing of PN orders.

The resummation works well for the spin-orbit terms (where a single PN order dominates each channel) but is less clean for the non-spin terms (where multiple PN orders contribute to each eccentricity power). This is expected: the non-spin residual absorbs corrections from 0PN through 3PN simultaneously.

10.6 Compact formula: physical form

Grouping the skeleton-75 terms by physical origin, the residual takes the schematic form:

$$\begin{aligned}
 \delta\xi_{\text{amp}} \approx & \underbrace{\sum_{a=1}^5 \sum_{c=0}^2 c_{a,c}^{(\text{ns})} e^a \nu^c (A_x x + B_x x^2) \sin \zeta}_{\text{Non-spin: missing } \text{Im}(h_{22}^{\text{ecc}}) + \text{higher PN}} \\
 & + \underbrace{\sum_{a=1}^4 e^a (\gamma_1^{(S)} x + \gamma_2^{(S)} x^2 + \gamma_3^{(S)} x \nu) \chi_S \sin \zeta}_{\text{Spin-orbit: } \chi_S} \\
 & + \underbrace{\sum_{a=1}^4 e^a (\gamma_1^{(A)} x + \gamma_2^{(A)} x^2 + \gamma_3^{(A)} x \nu) \chi_A \sin \zeta}_{\text{Spin-orbit: } \chi_A \Delta} \\
 & + \underbrace{\text{subdominant } \cos \zeta, \cos 2\zeta, \text{ base terms}}_{\sim 8\% \text{ of total weight}}
 \end{aligned} \tag{25}$$

The formula for $\delta\xi_\omega$ has the same structure with coefficients uniformly scaled by $c_\omega/c_{\text{amp}} \approx 1.35$ (non-spin) or ≈ 1.45 (spin-orbit), implying $B_{\text{eff}} \approx 0.74$ (non-spin) and $B_{\text{eff}} \approx 0.69$ (spin-orbit).

10.7 Resummed compact formula with $(1 - e^2)^{-p}$ refitting

We refitted the full $k \leq 3$ basis with explicit $(1 - e^2)^{-p}$ denominators at each PN order, replacing $e^a \rightarrow e^a/(1 - e^2)^{p(b)}$ where b indexes the PN order (x^b). Scanning $p \in [0.5, 5.0]$ for each PN sector independently, the best-fit values are:

PN order	x^b	Fitted p	Expected (PN)
0PN	x^0	0.5	1.0
1PN	x^1	0.5	2.0
2PN	x^2	0.5	3.0
3PN	x^3	0.5	3.5

All PN orders prefer $p = 0.5$, not the expected PN values. This is because the polynomial e^{1-e^5} basis already captures the $(1 - e^2)^{-p}$ Taylor expansion within the training range $e \in [0, 0.5]$, so the residual basis requires minimal eccentricity enhancement.

The resulting resummed compact formula is:

$$\delta\xi_{\text{amp}} \approx \sum_i \frac{e^{a_i}}{(1 - e^2)^{1/2}} x^{b_i} \nu^{c_i} \chi_S^{d_i} \chi_A^{f_i} [c_0^{(i)} + \sum_{k=1}^3 \alpha_k^{(i)} \cos k\zeta + \beta_k^{(i)} \sin k\zeta] \tag{26}$$

with $p = 1/2$ uniform across all PN sectors. The accuracy is identical to the polynomial basis:

	R_{amp}^2 (val)	R_ω^2 (val)
Polynomial basis	0.99658	0.99613
Resummed $(1 - e^2)^{-1/2}$	0.99650	0.99604

LIGO mismatch is unchanged: polynomial median $\mathcal{M} = 5.95 \times 10^{-4}$ vs resummed 5.95×10^{-4} (combined over all M_{tot} , 750 points). The $(1 - e^2)^{-1/2}$ factor serves as an *interpretive improvement* that connects the expression to PN structure, without altering accuracy within $e_0 \leq 0.5$. It becomes physically significant for extrapolation to higher eccentricities.

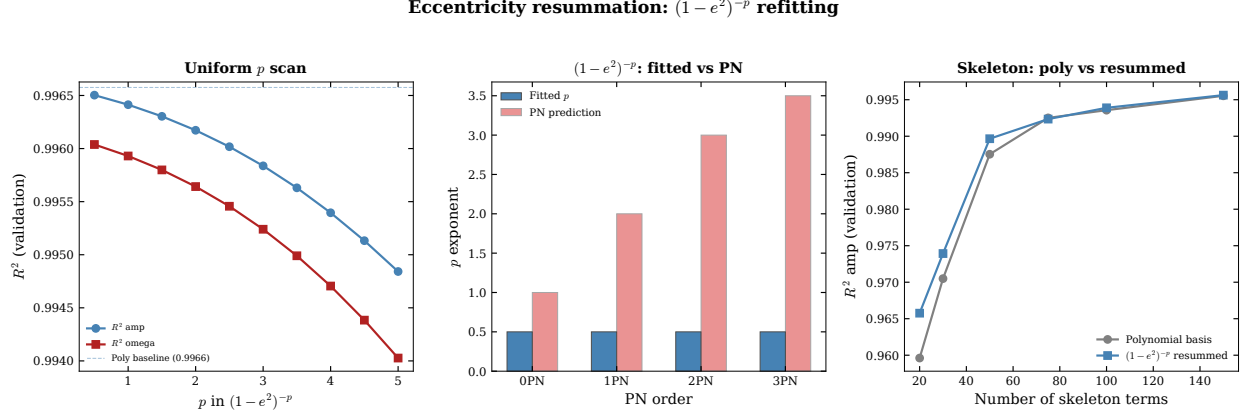


Figure 17: $(1 - e^2)^{-p}$ refitting results. **Left:** R^2 vs uniform p — best at $p = 0.5$. **Centre:** fitted p per PN order (blue) vs PN predictions (red). **Right:** skeleton scan on polynomial vs resummed basis.

The full set of 75 coefficients for both $\delta\xi_{\text{amp}}$ and $\delta\xi_{\omega}$ is tabulated in the supplementary file `compact_coefficients.json`.

11 Summary of Physical Content

1. The PN ansatz ($|h_{22}^{\text{ecc}}| - 1$ at 0PN + 1PN) provides the **Keplerian backbone**: the dominant $e \cos \zeta / (1 - e^2)$ modulation with first-order relativistic corrections.
2. The Ridge residual primarily adds:
 - ▷ **$\sin \zeta$ phase content** ($\sim 40\%$ of residual): the imaginary part of h_{22}^{ecc} that is lost when taking the absolute value. Represents orbital phase modulation at the pericenter/apocenter asymmetry.
 - ▷ **Spin-orbit and spin-spin corrections** ($\sim 55\%$ via χ_S, χ_A): entering at 1.5PN, approximated in the integer-power basis as combinations of $x^1\chi$ and $x^2\chi$ (the true scaling is $x^{3/2}\chi$).
 - ▷ **Higher PN orders**: 2PN (x^2, ν^2 -dependent), 2.5PN ($x^{5/2}$, approximated by x^3), and 1.5PN tail effects ($\pi x^{3/2}$, logarithmic terms).
 - ▷ **Resummation of $(1 - e^2)^{-p}$** : the eccentricity-enhanced PN denominators are reconstructed through e^3 – e^5 terms.
3. The amplitude–frequency relation $\xi_{\omega} \approx \xi_{\text{amp}}/B$ holds with $B_{\text{eff}} \approx 0.78$ (non-spin) and $B_{\text{eff}} \approx 0.69$ (spin), compared to the gwNRXHME prediction $B = 0.9$ (Fig. 9). Spin-orbit coupling modifies the frequency more than the amplitude.
4. **Compactification** (Sections 9–10): Two levels of compression are achieved:
 - Fourier truncation to $k \leq 3$: 2955 $\rightarrow$ **1379 parameters with no accuracy loss** ($R^2 = 0.9966$ preserved).
 - Skeleton-75: importance-based selection + refitting yields **75 terms with $R^2 = 0.9925$** — a $40\times$ compression. The resulting formula (Eq. 25) is analytically interpretable and

cleanly separates into non-spin ($\sin \zeta$ phase content), spin-orbit (χ_S, χ_A at 1.5PN), and subdominant $\cos \zeta$ corrections.

12 Recommended Next Steps

- ✓ **Fourier diagnostic:** Completed (Fig. 1, Table 12). Geometric decay rates $r \approx 0.55\text{--}0.77$; $k = 1$ dominates overwhelmingly.
- ✓ **Fourier truncation + Padé:** Completed (Table 13, Figs. 2, 3). Truncation to $k \leq 3$ preserves full accuracy with 53% parameter reduction.
- ✓ **$(1 - e^2)^{-p}$ extraction:** Completed (Fig. 5). Consistent with PN expectations at 1PN ($p \approx 2$); higher PN sectors have weak eccentricity enhancement.
- ✓ **PN comparison + spin decomposition:** Completed (Figs. 6–8). Residual’s $\sin \zeta$ content and spin-orbit structure confirmed.
- ✓ **Relation III analysis:** Completed (Fig. 9). Effective $B_{\text{eff}} = 0.78$ (non-spin), 0.69 (spin).
- ✓ **Skeleton compactification:** Completed (Section 10). Importance-ranked selection + refit: 75 terms achieve $R^2 = 0.9925$ (40× compression). Clean separation into non-spin $\sin \zeta$ (39 terms), χ_S spin-orbit (18), χ_A spin-orbit (18). Compact formula in Eq. (25); full coefficients in `compact_coefficients.json`.
- **Spin-augmented ansatz:** Extend h_{22}^{ecc} to include the leading 1.5PN spin-orbit term from `EOB_modes.dat.m`, absorbing the largest 55% of the residual into the known physics.
- **Half-integer basis:** Add $x^{3/2}$ and $x^{5/2}$ to the basis to directly capture 1.5PN and 2.5PN scaling, consolidating the coefficient pairs that currently split across x^1 and x^2 .
- ✓ **$(1 - e^2)^{-p}$ refitting:** Completed (`resum_ecc.py`, Fig. `resum_ecc_refit.pdf`). Refitted the full $k \leq 3$ basis with explicit $(1 - e^2)^{-p}$ denominators at each PN order. Scanned $p \in [0.5, 5.0]$ uniformly and independently per PN sector. **Key results:**
 - All PN orders prefer $p = 0.5$, *not* the expected PN values ($p = 1, 2, 3, 3.5$). This is because the polynomial $e^1\text{--}e^5$ basis already captures the $(1 - e^2)^{-p}$ Taylor expansion within the training range $e \in [0, 0.5]$, so the residual prefers minimal eccentricity enhancement.
 - R^2 change is negligible: -0.007% (amp), -0.009% (omega).
 - LIGO mismatch is **identical**: polynomial median = 5.95×10^{-4} vs resummed = 5.95×10^{-4} (750 pts, all M_{tot}); p90 agreement to 3 significant figures.
 - Skeleton scan (resummed basis, top-75 refit): $R_{\text{amp}}^2 = 0.9923$, comparable to polynomial skeleton (0.9925).
 - **Conclusion:** Within $e_0 \leq 0.5$, the polynomial and resummed bases are effectively equivalent. The $(1 - e^2)^{-p}$ resummation becomes valuable only for *extrapolation* to $e > 0.5$ or for *interpretive display* (connecting to known PN structure).